Residual Family Head: Making the Deployed Surrogate Extensible

Additive residual on a frozen backbone — and how far a cheap per-family adapter goes

Mie Spray Postprocessing Project

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The thread of this deck

We take the deployed Stage-3 surrogate from Chapter 5 and ask one question:

can we make it extensible to new injector families ***without*** *giving up accuracy — and how far does the cheap adapter reach?*

One principle ties every experiment together:

$$\boxed{\text{prediction} = \underbrace{\text{shared backbone}}_{\text{frozen, hot-started}} + \underbrace{\text{small per-family residual}}_{\text{trained, zero-initialised}}}$$

1. **Where Chapter 5 leaves off** — two open gaps.
2. **Theory** — additive residual on a frozen backbone.
3. **In-distribution climb** — residual head → FiLM → multi-task SVGP.
4. **Robustness** — LONO transfer and dataset-sensitivity checks.
5. **Boundary** — where the cheap adapter stops working.
6. **Significance** — what this changes about the thesis conclusions.

Roadmap

Where the thesis leaves the surrogate

Chapter 5 reports a three-stage MLP screening surrogate and an SVGP alternative backbone on the **conservative uncensored CDF table** (542,565 points). This is the *same table* the residual study uses — anchored by the SVGP, which scores **4.193 mm** in both documents.

Model (uncensored CDF, 542k pts)	**RMSE [mm]**	**P50 RMSE [mm]**
Calibrated Hiroyasu–Arai (physics)	10.261	–
Naber–Siebers w/ delay (physics)	9.286	–
Three-stage MLP, selected surrogate (5-seed)	4.265	2.848
Stage-3 SVGP (thesis' strongest predictor)	**4.193**	**2.649**

The thesis verdict is deliberately *not* a single winner: the SVGP wins point accuracy and LONO transfer; the MLP is kept as a conservative in-domain surrogate with an auditable workflow and an injection-onset head. LONO is fragile on **Nozzle0**, the only first-generation injector — a cross-design-family extrapolation failure.

Two open problems the thesis does not close

1. Deployment / extensibility

The deployed model conditions on injector family. To add a *new* family it needs either a hard-coded fallback identity or a freshly trained head. There is no clean, structural answer for an unseen injector.

2. Accuracy frontier

The best single model on the primary table is the SVGP at 4.193 mm. Can the same censoring-aware protocol be pushed further *without* abandoning the deployed backbone?

Claim of this study. Converting the deployed family-aware Stage-3 model into a *residual family head*

$$\mu(x, f) = \mu_{\mathrm{shared}}(h(x)) + \Delta_f(h(x))$$

addresses *both*: it gives a structural new-family fallback **and**, on the SVGP backbone, sets a new best on the primary table — while every variant hot-starts from the deployed checkpoint and freezes its trunk.

Roadmap

Conditioning on injector family: three designs

Let $h(x)$ be the shared trunk representation of the low-dimensional condition descriptor x (time, A-scaled amplitude, log-pressure residual). Family f can enter the mean in three ways:

Design	Family signal	New family	CDF RMSE
One-hot input	feature into one shared head $\mu(h, f)$	implicit	4.265
Per-family head	each family owns a decoder $\mu_f(h)$	needs fallback / trained head	5.056
Residual head	shared mean + small residual $\mu_{\text{shared}} + \Delta_f$	zero residual → shared	4.631

- The deployed **per-family head** buys per-family capacity but fragments the data — aggregate RMSE rises to 5.056 vs the single-head surrogate's 4.265 — *and* leaves no answer for an unseen family.
- The **residual head** recovers most of that lost accuracy (5.056→4.631) while making the fallback *structural*.

Residual head: identity warm-start + structural fallback

$$\mu(x, f) = \mu_{\text{shared}}(h(x)) + \Delta_f(h(x)), \qquad \Delta_f \equiv 0 \text{ at init}$$

Residual-learning view. Like a residual block, each family learns a *correction* to a strong shared predictor, not the whole map. Zero-initialised Δ_f means the model *equals* the deployed production model at step 0; training only perturbs it.

Structural fallback. An unseen family carries $\Delta_f \equiv 0$, so it *naturally* predicts μ_{shared} — no hard-coded `fallback_family_id`.

Shrinkage prior. The Δ_f weight-decay (`delta_L2`) is an explicit prior pulling each family back toward the shared model — hierarchical shrinkage trading per-family fit against shared-model proximity.

Parameter-efficient transfer

Freeze the backbone, train only tiny per-family modules:

- trunk frozen: 334,720 params
- shared mean frozen: 16,641 params
- trainable: 49,923 (Δ heads + log-var)

The adapter / PEFT paradigm: a frozen backbone plus tiny task heads, so a new family is a new adapter — not a retrain.

Warm-start mapping

copy `trunk.*`; copy `log_var_head.*`; `mu_heads.1`→`mu_shared_head`; `delta_heads.*` zero-init, then 10-epoch KD mimic.

Two richer instantiations of the same principle

FiLM: condition the *representation*, not just the offset

$$h'_l = \gamma_{f,l} \odot h_l + \beta_{f,l}, \qquad \mu(x, f) = \mu_{\text{shared}}(h'_L) + \Delta_f(h'_L)$$

Feature-wise linear modulation inserts a per-family affine transform *inside* the trunk. Identity init $(\gamma = 1, \beta = 0, \Delta_f = 0)$ preserves the exact warm-start. It asks: does injector family change *how* $(a, \Delta p)$ are represented, or only the final mean offset?

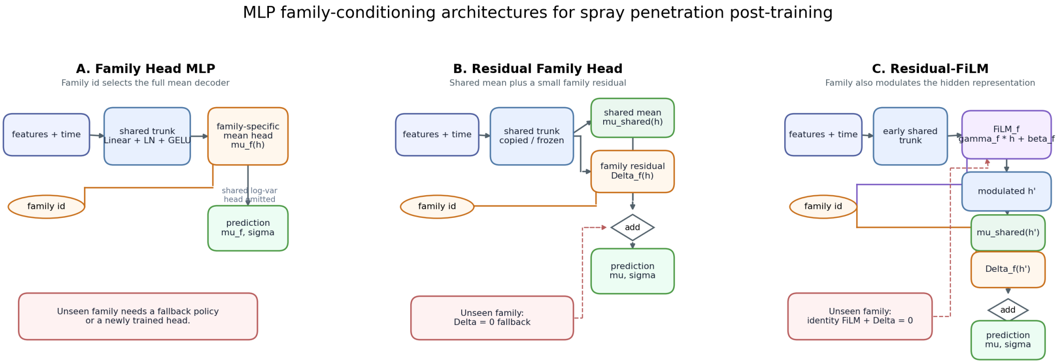
Additive-residual multi-task SVGP: the same idea on a GP backbone

$$g(x, f) = g_{\text{shared}}(x) + \delta_f(x)$$

A shared sparse variational GP hot-started from the deployed Stage-3 SVGP, plus a per-family residual GP. Family id *routes* the residual but stays out of the kernel feature vector; an unseen family falls back to g_{shared}. This is a shared-plus-per-family *additive (hierarchical)* multi-task GP; its probabilistic backbone yields the best-calibrated residual variant in this study (ECE 0.026).

All three share the contract: **hot-start the deployed checkpoint, freeze the backbone, train only zero-/identity-initialised per-family residuals.**

Architecture: Family Head vs Residual vs FiLM



Left: per-family decoders. Middle: shared mean + zero-init residual. Right: identity FiLM inside the trunk + residual head. Green modules are the only trainable parameters; everything grey is hot-started and frozen.

Experiment contract

Fixed production baseline

- Baseline run: `distill_cdf_family_head_v3_FROZEN_anchor_off_20260529_213346` (the deployed per-family-head model, CDF RMSE 5.056).
- Feature contract: `a_dp050_plus_pressures`; device CUDA; seed inherited from the production run.
- Evaluation: canonical point tables (`cdf_uncensored`, `p50_observed`, `q1_grid_all`) — `cdf_uncensored` is the thesis' 542k-point primary table.

Knob	**Values**
Residual shrinkage	`0`, `1e-4`, `1e-3`, `1e-2`
Mimic warm-up	10 epochs; teacher is the deployed production model
Refinement	150 max epochs, patience 20, trunk frozen
Trainable params	49,923: residual deltas + shared log-var head

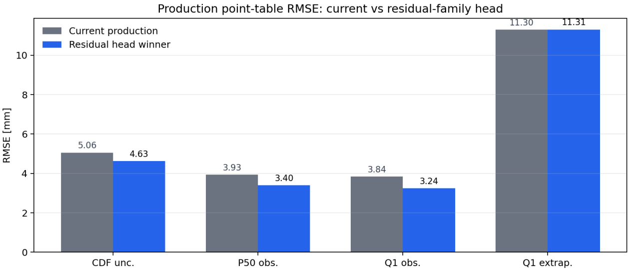
Roadmap

Training log read

delta L2	mimic epochs	stop epoch	test loss	test delta RMS
0	10	104	2.074	0.816
1e-4	10	44	2.085	0.818
1e-3	10	110	2.070	0.813
1e-2	10	84	2.083	0.810

- All four runs completed and early-stopped cleanly.
- The mimic stage reduced the initial production-head mismatch before raw refinement.
- Delta magnitudes stayed tightly clustered ($\mathrm{RMS} \approx 0.81$ in A-scaled space), so the shrinkage sweep mainly changes calibration and tail behaviour, not gross residual size.

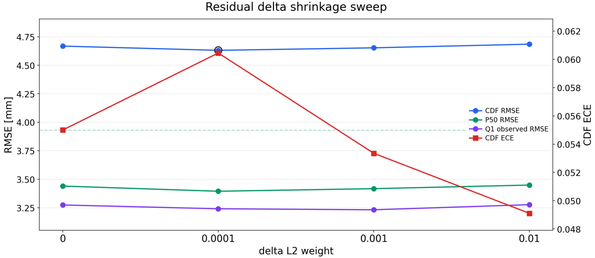
Step 1 — Residual head beats the deployed family head

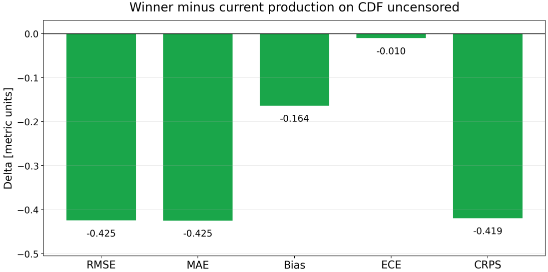


Model	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 obs RMSE
Deployed family head	5.056	0.0708	2.801	3.931	3.839
Residual winner (1e-4)	**4.631**	**0.0604**	**2.382**	**3.395**	**3.242**

CDF RMSE -0.425 mm, bias toward zero $(+0.343 \to +0.180)$, calibration (ECE $0.0708 \to 0.0604$, CRPS $2.801 \to 2.382$).

Step 1 — shrinkage sweep and per-point deltas





- *All* residual settings beat the deployed family head on CDF RMSE; `1e−4` gives the best primary RMSE (4.631 mm).
- Stronger shrinkage improves ECE and extrapolated Q1 but gives back a little primary RMSE — the sweep is a calibration/tail knob.

Step 2 — FiLM: the family signal is partly representational

The recommended MLP-only ablation, run on production (no LONO): an identity FiLM adapter after the final trunk block, hot-started from the residual family head, trunk and shared mean frozen.

Model	film L2	delta L2	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 obs RMSE
Source residual FH	–	1e-2	4.685	0.0491	2.368	3.449	3.277
Residual-FiLM	1e-4	1e-3	**4.510**	**0.0404**	**2.242**	**3.166**	**3.110**

- CDF RMSE -0.176 mm; P50 -0.283 mm; ECE $0.0491 \rightarrow 0.0404$, CRPS $2.368 \rightarrow 2.242$; Q1 extrapolated effectively flat $(11.248 \rightarrow 11.265)$.
- All nine `(film,delta)` settings beat the source residual head: family difference is useful as a hidden-representation modulation, not only as an output offset.

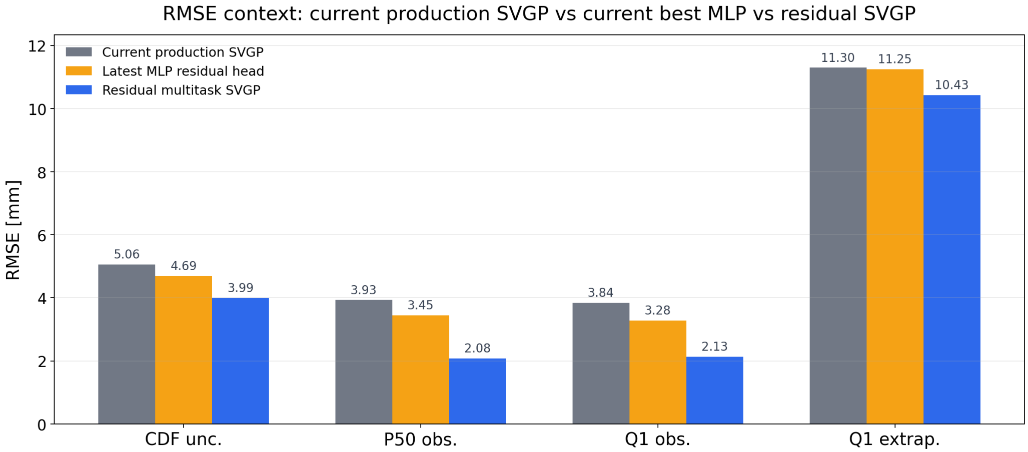
Step 3 — Additive-residual multi-task SVGP: new best

The same additive-residual idea on the probabilistic backbone: a shared SVGP hot-started from the deployed Stage-3 checkpoint + per-family residual GPs, family id outside the kernel, zero-residual fallback for unseen families.

Model	**delta L2**	**CDF RMSE**	**CDF ECE**	**CDF CRPS**	**P50 RMSE**	**Q1 extrap RMSE**
MLP residual FH	1e−2	4.685	0.0491	2.368	3.449	11.248
Residual SVGP	1e−4	**3.987**	**0.0262**	**2.040**	**2.077**	**10.429**

- Beats the single-output SVGP baseline (CDF $4.193 \rightarrow 3.987$) — and the probabilistic backbone halves ECE vs the residual MLP (0.026 vs 0.060).
- A narrow, stable shrinkage sweep: 1e−4 is the sweet spot for primary RMSE *and* calibration on the evaluated tables.

Step 3 — Residual SVGP vs current best



The climb on one table (uncensored CDF, lv2 542k points)

Model	**CDF RMSE**	**P50 RMSE**	**Origin**
Naber–Siebers (physics)	9.286	–	thesis
Three-stage MLP, selected surrogate	4.265	2.848	thesis
Stage-3 SVGP	4.193	2.649	thesis (prev. best)
Deployed family head	5.010	3.907	deployment variant
Residual FH (1e-4)	4.631	3.395	**this work**
Residual-FiLM	4.510	3.166	**this work**
Residual SVGP (1e-4)	**3.987**	**2.077**	**this work (new best)**
Residual SVGP, QC-gated retrain	3.922	1.912	sensitivity re-score

Read honestly: the residual *MLP* (4.631) recovers most of the accuracy the per-family head gave up (5.010) but does *not* beat the simpler single-head surrogate (4.265). The structural accuracy frontier is advanced by the residual *SVGP* (3.987), which beats the thesis' strongest model (4.193). The QC-gated retrain improves the residual SVGP again on the same lv2 root (3.987→3.922) but is treated as a sensitivity result, not the sole proof.

Roadmap

LONO: does the residual win survive an unseen nozzle?

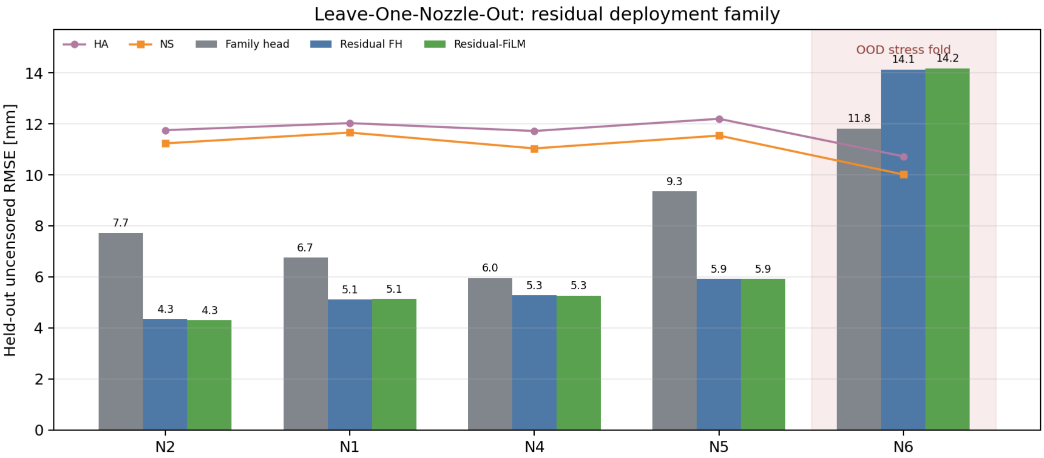
Protocol

- Hold out Nozzle 1, 2, 4, 5, 6 one at a time; **Nozzle0 stays in train** as the sole first-generation anchor (holding it out is a separate OOD study).
- Each fold trains Stage-1→2→3 *from scratch* with the nozzle removed — no production-checkpoint leakage.
- Arms: family-head teacher, residual FH (`1e-4`), residual-FiLM (`film 1e-4, delta 1e-3`).

Aggregation	Family head	Residual FH	Residual-FiLM	HA / NS
All 5 folds	8.32 ± 2.33	6.95 ± 4.05	6.96 ± 4.08	11.69 / 11.10
4 folds excl. N6	7.44 ± 1.46	$\mathbf{5.16 \pm 0.65}$	$\mathbf{5.15 \pm 0.67}$	11.93 / 11.37

On in-family folds the residual conversion cuts family-head RMSE by 31% (7.44→5.16) and nearly halves the fold spread. N6 is reported as an OOD stress fold, not hidden in the mean. (The thesis LONO holds out N0,1,2,4,5; here N0 stays in as the family anchor and N6 is the in-protocol sparse/OOD fold — N6 is not a thesis fold.)

LONO: per-fold read



Residual FH improves every in-family fold: N2 -3.37, N1 -1.64, N4 -0.68, N5 -3.43 mm. N6 $+2.31$ (regression).

LONO: verdict

- **The residual deployment win generalises (MLP arms).** On all four in-family held-out nozzles the residual arms beat the family-head teacher and dominate the HA/NS physics baselines on nozzles they never trained on. (LONO here covers the MLP variants; the residual SVGP is not yet LONO-tested.)
- **Bias repair.** The family head over-predicts ($+3$ to $+8$ mm); the residual conversion moves the 4-fold mean bias from $+5.07$ to $+0.78$ mm and restores coverage (0.44→0.78 on N5).
- **FH $\approx$ FiLM under LONO.** Per-fold numbers agree within 0.04 mm. FiLM's in-distribution edge does *not* translate into unseen-nozzle transfer — for a new nozzle, the cheap output residual is as good as the representation adapter.
- **N6 is a true stress fold.** All MLP arms collapse to 12–14 mm and the residual arms under-predict severely (bias -10.5). This foreshadows the Nozzle0 case and motivates a separate OOD study, not another adapter sweep.

Dataset sensitivity matrix: explicit lv2 vs lv3 roots

Model	**Train root**	**lv2 CDF**	**lv3 CDF**	**lv2 P50**	**lv3 P50**
Family head baseline	pre-QC	5.010	4.901	3.907	3.785
Residual FH	pre-QC	4.631	4.519	3.395	3.264
Residual-FiLM	pre-QC	4.510	4.401	3.166	3.035
Residual SVGP	pre-QC	3.987	3.921	2.077	2.026
Residual SVGP	lv3 QC	**3.922**	**3.848**	**1.912**	**1.829**

Controlled read: every row is re-scored through an explicit `--synthetic-root`; `MLP/synthetic_data` is never used. The lv3 root is a target/evaluation cleanup sensitivity check, so its lower RMSE is evidence of robustness to cleaning choices, not a replacement for the lv2 primary comparison.

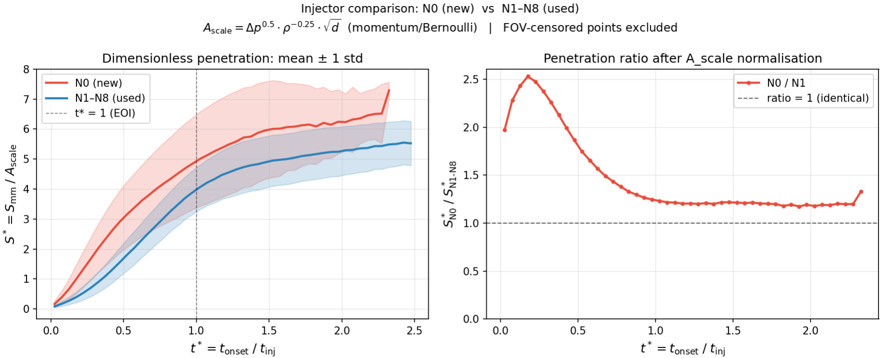
QC-gated retrain: cleanup vs model gain

Model	Cleanup old lv3 – old lv2	QC retrain QC lv3 – old lv3	Net QC lv3 – old lv2
Residual FH	-0.112	+0.071	-0.041
Residual-FiLM	-0.108	-0.001	-0.109
Residual SVGP	-0.067	-0.073	-0.140

- **Post-training is the main structural move:** on lv2, family head $5.010 \rightarrow$ residual FH 4.631 $\rightarrow$ FiLM 4.510 $\rightarrow$ residual SVGP 3.987.
- **QC retraining is not a blanket MLP win:** FH regresses and FiLM is flat. The clear extra retrain gain is the residual SVGP (CDF -0.073, P50 -0.197, q1-observed -0.170).
- **The extrapolated tail barely moves** for residual SVGP (q1-extrapolated -0.021); QC mainly improves the representative-window fit rather than changing the far continuation.

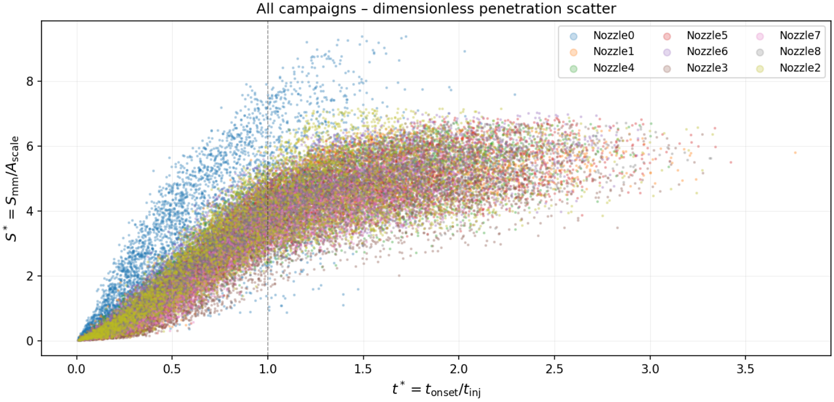
Roadmap

Evidence: Nozzle0 is not just another used injector



- Even after momentum/Bernoulli-style normalisation, N0 remains distinct: early-time response reaches $\sim 2.5\times$ N1–N8 and stays above 1.
- N0 is a **cross-injector-distribution** problem, not merely a missing per-family output head.

Evidence: the full scatter separates N0's early branch



Nozzle0 occupies a higher early-time envelope before and around $t^* = 1$, while the used injectors mostly overlap. The residual head can remove a family offset, but it cannot create a trunk representation for a branch the frozen backbone never saw.

N6 and N0: two faces of out-of-design-family

The residual head's structural fallback works *when the new family is in design family*. Two cases mark the boundary:

Nozzle6 (LONO stress fold)

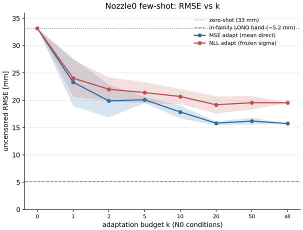
Data-sparsest fold (20k points). All MLP arms collapse to 12–14 mm; the residual arms are 2.3 mm worse than the family head and under-predict (bias -10.5). Specialisation *over-commits* on a family the trunk barely supports.

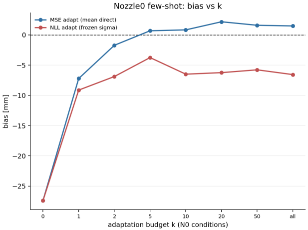
Nozzle0 (the genuine new injector)

The only first-generation (BC2022) injector — same orifice geometry, empirically distinct signature. Zero-shot (trunk never saw N0): 33.16 mm. The deployment question: how many N0 conditions does the per-family delta head need to recover?

Both echo the thesis' Nozzle0 cross-design-family failure: a per-family *output* correction cannot synthesise a representation the frozen trunk never learned.

N0 few-shot adaptation: the limit, quantified





Freeze everything except `delta_heads[0]`; fine-tune on k N0 conditions; evaluate uncensored RMSE (in-family LONO band ≈5 mm).

k	NLL	MSE
0 (zero-shot)	33.16	33.16
2	22.01	19.88
10	20.68	17.85
20	19.17	15.81
all (203)	19.54	**15.73**

1. **The frozen variance head throttles NLL adaptation.** On OOD N0 the frozen log-var emits a huge σ (mean std ∼1443 mm at $k{=}0$), down-weighting the mean gradient. Fitting the mean directly (MSE) drops the floor by ∼4 mm and removes the bias by $k{=}2$.
2. **Even unthrottled, delta-only adaptation plateaus at ∼15.7 mm —** far above the ∼5 mm in-family band. A per-family offset removes the global bias but cannot close the residual spread: N0 needs trunk-level capacity (unfreeze / FiLM-all-blocks / retrain), not a new output head.

Roadmap

What this adds to the thesis

1. **A new best on the primary table.** The residual multi-task SVGP reaches 3.987 mm CDF and 2.077 mm P50 on the same uncensored table — beating the thesis' strongest model (SVGP 4.193 / P50 2.649) by ∼5% on CDF RMSE and 0.57 mm on P50. QC-gated retraining still improves when re-scored on lv2 (3.922 / 1.912) and reaches 3.848 / 1.829 on lv3; the latter is sensitivity evidence, not the sole final proof.
2. **The deployment gap is closed structurally — within design family.** A new injector family is a zero-initialised residual that *automatically* falls back to the shared model — no hard-coded fallback identity, no retrain, ∼50k trainable params on a frozen backbone. The fallback is mechanically clean for *any* family but predictively safe only in-design-family (the N0/N6 boundary).
3. **The MLP branch's LONO weakness is partly repaired.** The residual conversion turns the family head's $+5$ mm over-prediction into $+0.78$ mm and cuts in-family LONO RMSE by 31% — directly strengthening Chapter 5's transfer story for the MLP stack.
4. **The boundary is drawn honestly.** N6/N0 show the cheap adapter recovers in-design-family injectors but not out-of-design-family ones — a constructive, quantified refinement of the thesis' Nozzle0 failure rather than a hidden caveat.

Verdict and next steps

Verdict

The single principle — *additive residual on a hot-started, frozen backbone* — upgrades the deployed surrogate along both axes the thesis left open: it makes the model cleanly extensible to new families *and*, on the SVGP backbone, sets a new accuracy/calibration best on the primary uncensored CDF table, all while preserving the auditable deployment path.

- Package the residual SVGP as the candidate deployed model, with the QC-gated checkpoint as the current strongest sensitivity row; keep residual-FiLM as the best MLP-only option and residual FH as the simplest fallback. *Caveat:* LONO transfer is validated for the MLP arms only; the residual SVGP's LONO behaviour is unmeasured here, though the thesis SVGP already out-transfers the MLP (5.95 vs 7.00 mm excl. N0) — confirm with a residual-SVGP LONO fold.
- For genuinely new injectors: adapt the log-var head alongside the delta, and test trunk-level capacity (FiLM-all-blocks / last-block unfreeze) to close the N0/N6 residual spread.

Artifacts: `residual_from_prod_eval_summary.csv`; `residual_svgp_vs_latest_mlp.csv`; `lono_modified_only/`; `dataset_ablation_qc_gated/`; `cross_dataset_lv2_lv3/`; `n0_fewshot_adaptation/`; figures under `figs/`.